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Areas of Expertise

Computer Vision, Deep Learning, Remote Sensing, Medical Imaging, Multimodal Integration

Contents

1	Education	1
2	Appointments and Affiliations	1
3	Awards	3
4	Publications	3
5	Talks	19
6	Service	23
7	Teaching	28
8	Lab Members / Mentees	29

1 Education

2005–2010	Ph.D. in Computer Science Adviser: Robert Pless, Ph.D. Thesis: Calibrating and Using the Global Network of Outdoor Webcams	Washington University in St. Louis
1995–1999	B.S. in Computer Science (Minor in Mathematics) <i>Summa Cum Laude</i> with Honors	University of Missouri

2 Appointments and Affiliations

Active

Vice Provost for Artificial Intelligence May 2026–	Washington University St. Louis, MO
Professor (with Tenure) 2022–	Dept. of Computer Science & Engineering, Washington University St. Louis, MO
Affiliated Faculty 2022–	Division of Computational & Data Sciences, Washington University St. Louis, MO

Affiliated Faculty 2022–	Imaging Science Program, Washington University <i>St. Louis, MO</i>
Affiliated Faculty 2023–	Electrical and Systems Engineering, Washington University <i>St. Louis, MO</i>
Affiliated Faculty 2023–	AI for Health Institute, Washington University <i>St. Louis, MO</i>
Associate Faculty 2022–	Taylor Geospatial Institute <i>St. Louis, MO</i>
Faculty Scholar 2023–	Center for the Environment, Washington University <i>St. Louis, MO</i>
Biodiversity Fellow 2022–	Living Earth Collaborative <i>St. Louis, MO</i>
Technical Consultant / Owner / Founder 2019–present	Multidomain Vision Research, LLC <i>St. Louis, MO</i>

Prior

Assistant Vice Provost for Digital Transformation Jan 2025–May 2026	Washington University <i>St. Louis, MO</i>
Professor (with Tenure) 2021–2022	Dept. of Computer Science, University of Kentucky <i>Lexington, KY</i>
Director of Graduate Studies (Data Science) 2020–2022	Dept. of Computer Science, University of Kentucky <i>Lexington, KY</i>
Member 2017–2022	Institute for Biomedical Informatics, University of Kentucky <i>Lexington, KY</i>
Associate Professor (with Tenure) 2016–2021	Dept. of Computer Science, University of Kentucky <i>Lexington, KY</i>
Co-Department Chair (interim) 2019–2020	Dept. of Computer Science, University of Kentucky <i>Lexington, KY</i>
Affiliated Faculty 2010–2019	Center for Visualization and Virtual Environments, University of Kentucky <i>Lexington, KY</i>
Visiting Research Scientist (sabbatical) 2017–2018	Orbital Insight, Inc. <i>Mountain View, CA</i>
Assistant Professor 2010–2016	Dept. of Computer Science, University of Kentucky <i>Lexington, KY</i>
Computer Vision Research Intern 2008	ObjectVideo, Inc. <i>Reston, VA</i>
Graduate Research Assistant 2005–2010	Dept. of Computer Science & Engineering, Washington University <i>St. Louis, MO</i>

3 Awards

- Best Paper Award (out of 75 valid submissions) [EarthVision Workshop 2024 at IEEE/CVF Computer Vision and Pattern Recognition (CVPR)]
- Highlighted Reviewer Recognition (top 8%) [ICLR 2022]
- Outstanding Reviewer Recognition [BMVC 2021]
- Outstanding Reviewer Recognition (top 10%) [NeurIPS 2020]
- Outstanding Reviewer Recognition [ICCV 2019]
- University of Kentucky, College of Engineering Dean’s Award for Excellence in Research [2018]
- Google Faculty Research Award [2018]
- Outstanding Reviewer Recognition [CVPR 2017]
- National Science Foundation CAREER Award [2016]
- Google Faculty Research Award [2016]
- Best Student Paper Award at Applied Imagery Pattern Recognition [2009]
- Ph.D. Forum Prize at the ACM/IEEE International Conference on Distributed Smart Cameras [2009]
- Best Talk Award for the Doctoral Student Seminar, Department of Computer Science, the Washington University in St. Louis, [Fall 2006]

4 Publications

Preprints

- [1] S. Sastry, D. Cher, B. Wei, A. Dhakal, S. Khanal, D. Gupta, and N. Jacobs, “TerraDiT: Point-conditioned diffusion transformer for satellite image synthesis,” in *TerraBytes — Towards global datasets and models for Earth observation (European Conference on Computer Vision (ECCV) Workshop)*, Sep. 2026. arXiv: [2603.02172 \[cs.CV\]](#).
- [2] Y. Cui, F. Qiao, and N. Jacobs, *StereoGenBench: A synthetic multi-camera benchmark for stereo generation under controlled baseline regimes*, May 2026. arXiv: [2605.23237 \[cs.CV\]](#).
- [3] C. Robinson, G. Muhawenayo, S. Khanal, Z. Fang, I. Corley, A. M. Tárano, L. Estes, J. Marcus, N. Jacobs, H. Kerner, I. Becker-Reshef, and J. M. L. Ferres, *The first global agricultural field boundary map at 10m resolution*, May 2026. arXiv: [2605.11055 \[cs.CV\]](#).
- [4] H. J. Koch, J. Zhu, S. Bhatt, N. Jacobs, and J. M. Dortch, “Assessing deep learning segmentation models for mapping landslides from lidar-based elevation data,” Mar. 2026. DOI: [10.22541/essoar.177265426.62613754/v1](#) [Online]. Available: <http://dx.doi.org/10.22541/essoar.177265426.62613754/v1>
- [5] Y. Luo, F. Qiao, Z. Xiong, Y. Li, and N. Jacobs, *GenOpticalFlow: A generative approach to unsupervised optical flow learning*, Mar. 2026. arXiv: [2603.22270 \[cs.CV\]](#).
- [6] Z. Xiong, Y. Song, L. He, W. Xiong, Y. Yuan, F. Qiao, and N. Jacobs, *PhysAlign: Physics-coherent image-to-video generation through feature and 3d representation alignment*, Mar. 2026. arXiv: [2603.13770 \[cs.CV\]](#).
- [7] F. Qiao, Z. An, Z. Xiong, S. Belongie, and N. Jacobs, *Track2View: 4d-consistent camera-controlled video generation via paired 3d point tracks*, 2026. arXiv: [2606.15534 \[cs.CV\]](#).

- [8] Z. Xiong, Y. Song, H. Kang, Q. Yan, L. Jiang, J. Yang, Z. Fu, S. Fotiadis, A. Wang, Z. Liu, B. Liu, Y. Yang, X. Lu, and N. Jacobs, *ActWorld: From explorable to interactive world model via action-aware memory*, 2026. arXiv: [2606.17730 \[cs.CV\]](#).
- [9] C. Mosig, T. Kattenborn, D. Montero Loaiza, J. Vanja-Jehle, J. Brandt, N. Jacobs, S. Khanal, E. Xing, M. Schwartz, H. C. Muller-Landau, M. Beloiu, A. Bozzini, Y. Cheng, K. Ganz, B. Grüning, H. Hartmann, J. Hempel, S. Horion, S. Junttila, K. Korznikov, G. Kraemer, M. Mönks, D. Nardi, P. Neumeier, J. Schmid, S. Soltani, M. Therese-Schmehl, J. Veitch-Michaelis, and M. Mahecha, “Sub-pixel mapping of disturbance and tree mortality dynamics from sentinel-2 time series around the globe,” *Eartharxiv*, 2026. DOI: [10.31223/X5B18W](#) [Online]. Available: <https://eartharxiv.org/repository/view/11912/>
- [10] K. Klemmer, E. Rolf, M. Rußwurm, G. Camps-Valls, M. Czerkawski, S. Ermon, A. Francis, N. Jacobs, H. Kerner, L. Mackey, G. Mai, O. M. Aodha, M. Reichstein, C. Robinson, D. Rolnick, E. Shelhamer, V. Sitzmann, D. Tuia, and X. Zhu, “Earth embeddings: Toward artificial intelligence-centric representations of our planet,” *IEEE Geoscience and Remote Sensing Magazine*, pp. 2–15, 2026. DOI: [10.1109/MGRS.2026.3710416](#)

Refereed Conference Papers

- [1] S. Khanal, Y. Cui, D. Cher, E. Xing, B. Wei, S. Sastry, and N. Jacobs, “Genesis: A generative engine for hierarchical satellite image synthesis,” in *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (ACM SIGSPATIAL)*, Nov. 2026. DOI: [10.1145/3841645.3843313](#) arXiv: [2609.02683 \[cs.CV\]](#).
- [2] D. Cher, H. Iqbal, E. Xing, B. Wei, and N. Jacobs, “Tesselating the earth,” in *European Conference on Computer Vision (ECCV)*, Sep. 2026. arXiv: [2606.27514 \[cs.CV\]](#).
- [3] T. Gao, J. Lin, D. Beaulieu, and N. Jacobs, “Trajectory-aware cross-view geo-localization with sequential observations,” in *European Conference on Computer Vision (ECCV)*, Sep. 2026. arXiv: [2607.15491 \[cs.CV\]](#).
- [4] B. Wei, S. Sastry, D. Cher, E. Xing, and N. Jacobs, “TerraDiT-Ω: Unified spatial control for satellite image synthesis with any geospatial primitive,” in *European Conference on Computer Vision (ECCV)*, Sep. 2026. eprint: [2606.31029 \(cs.CV\)](#).
- [5] Z. Xiong, X. Ye, B. Yaman, S. Cheng, Y. Lu, J. Luo, N. Jacobs, and L. Ren, “UniDrive-WM: Unified understanding, planning and generation world model for autonomous driving,” in *European Conference on Computer Vision (ECCV)*, Sep. 2026. arXiv: [2601.04453 \[cs.CV\]](#).
- [6] A. Dhakal, S. Khanal, S. Sastry, J. Arndt, P. A. Dias, D. Lunga, and N. Jacobs, “SimLBR: Learning to detect fake images by learning to detect real images,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2026. arXiv: [2602.20412 \[cs.CV\]](#).
- [7] G. Muhawenayo, C. Robinson, S. Khanal, Z. Fang, I. Corley, A. Wollam, T. Gao, L. Strnad, R. Avery, L. Estes, A. M. Tárrano, N. Jacobs, and H. Kerner, “PRUE: A practical recipe for field boundary segmentation at scale,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2026.
- [8] S. Sastry, S. Khanal, A. Dhakal, J. Lin, D. Cher, P. Jarosz, and N. Jacobs, “ProM3E: Probabilistic masked multimodal embedding model for ecology,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2026. arXiv: [2511.02946 \[cs.CV\]](#).
- [9] A. Sarkar, S. Sastry, A. Pirinen, N. Jacobs, and Y. Vorobeychik, “DiffVAS: Diffusion-guided visual active search in partially observable environments,” in *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, (oral), May 2026. arXiv: [2605.15519 \[cs.CV\]](#).
- [10] D. Cher, B. Wei, S. Sastry, and N. Jacobs, “VectorSynth: Fine-grained satellite image synthesis with structured semantics,” in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Mar. 2026. arXiv: [2511.07744 \[cs.CV\]](#).

- [11] A. Wollam, K. Ashley, M. Shugaev, O. Arend, I. Y. Semenov, H. Dashtestani, S. Ravi, and N. Jacobs, “Towards unconstrained cross-view pose estimation,” in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Mar. 2026.
- [12] A. Elallaf, N. Jacobs, X. Ye, M. Chen, and G. Liang, “Beta distribution learning for reliable roadway crash risk assessment,” in *Association for the Advancement of Artificial Intelligence (AAAI)*, Jan. 2026. arXiv: [2511.04886 \[cs.CV\]](#).
- [13] E. Xing, A. Stylianou, R. Pless, and N. Jacobs, “QuARI: Query adaptive retrieval improvement,” in *Neural Information Processing Systems (NeurIPS)*, vol. 2505.21647, Dec. 2025. arXiv: [2505.21647 \[cs.CV\]](#).
- [14] F. Qiao, Z. Xiong, E. Xing, and N. Jacobs, “Towards open-world generation of stereo images and unsupervised matching,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, Oct. 2025. arXiv: [2503.12720 \[cs.CV\]](#).
- [15] S. Sastry, A. Dhakal, E. Xing, S. Khanal, and N. Jacobs, “Global and local entailment learning for natural world imagery,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, vol. 2506.21476, Oct. 2025. arXiv: [2506.21476 \[cs.CV\]](#).
- [16] A. Dhakal, S. Sastry, S. Khanal, A. Ahmad, E. Xing, and N. Jacobs, “RANGE: Retrieval augmented neural fields for multi-resolution geo-embeddings,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2025. arXiv: [2502.19781 \[cs.CV\]](#).
- [17] E. Xing, P. Kolouju, R. Pless, A. Stylianou, and N. Jacobs, “ConText-CIR: Learning from concepts in text for composed image retrieval,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2025. arXiv: [2505.20764 \[cs.CV\]](#).
- [18] H. R. Kerner, S. Chaudhari, A. Ghosh, C. Robinson, A. Ahmad, E. Choi, N. Jacobs, C. Holmes, M. Mohr, R. Dodhia, J. M. L. Ferres, and J. Marcus, “Fields of The World: A machine learning benchmark dataset for global agricultural field boundary segmentation,” in *Association for the Advancement of Artificial Intelligence (AAAI)*, vol. 2409.16252, Feb. 2025. arXiv: [2409.16252 \[cs.CV\]](#).
- [19] A. Sarkar, A. DiChristofano, S. Das, P. Fowler, N. Jacobs, and Y. Vorobeychik, “Active geospatial search for efficient tenant eviction outreach,” in *Association for the Advancement of Artificial Intelligence (AAAI)*, Feb. 2025. arXiv: [2412.17854 \[cs.LG\]](#).
- [20] S. Sastry, S. Khanal, A. Dhakal, A. Ahmad, and N. Jacobs, “TaxaBind: A unified embedding space for ecological applications,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Feb. 2025. arXiv: [2411.00683 \[cs.CV\]](#).
- [21] M. Lanier, Y. Xu, N. Jacobs, C. Zhang, and Y. Vorobeychik, “Learning interpretable policies in hindsight-observable POMDPs through partially supervised reinforcement learning,” in *IEEE International Conference on Machine Learning and Applications*, Dec. 2024. arXiv: [2402.09290 \[cs.LG\]](#).
- [22] A. Sarkar, S. Sastry, A. Pirinen, C. Zhang, N. Jacobs, and Y. Vorobeychik, “GOMAA-Geo: Goal modality agnostic active geo-localization,” in *Neural Information Processing Systems (NeurIPS)*, Dec. 2024. arXiv: [2406.01917 \[cs.CV\]](#).
- [23] S. Khanal, E. Xing, S. Sastry, A. Dhakal, Z. Xiong, A. Ahmad, and N. Jacobs, “PSM: Learning probabilistic embeddings for multi-scale zero-shot soundscape mapping,” in *ACM Multimedia*, Oct. 2024. DOI: [10.1145/3664647.3681620](#) arXiv: [2408.07050 \[cs.CV\]](#).
- [24] O. Skean, A. Dhakal, N. Jacobs, and L. G. S. Giraldo, “FroSSL: Frobenius norm minimization for self-supervised learning,” in *European Conference on Computer Vision (ECCV)*, Oct. 2024. arXiv: [2310.02903 \[cs.LG\]](#).
- [25] A. Sarkar, A. DiChristofano, S. Das, P. Fowler, N. Jacobs, and Y. Vorobeychik, “Geospatial active search for preventing evictions,” in *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, May 2024.
- [26] C. Greenwell, M. Leotta, J. Crall, N. Jacobs, M. Purri, K. Dana, A. Hadzic, and S. Workman, “Watch: Wide-area terrestrial change hypercube,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2024.

- [27] A. Sarkar, M. Lanier, S. Alfeld, J. Feng, R. Garnett, N. Jacobs, and Y. Vorobeychik, “A visual active search framework for geospatial exploration,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2024.
- [28] S. Sastry, S. Khanal, A. Dhakal, D. Huang, and N. Jacobs, “BirdSat: Cross-view contrastive masked autoencoders for bird species classification and mapping,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2024.
- [29] M. Shugaev, I. Semenov, K. Ashley, M. Klaczynski, N. Cuntoor, M. W. Lee, and N. Jacobs, “ArcGeo: Localizing limited field-of-view images using cross-view matching,” in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2024.
- [30] A. Sarkar, N. Jacobs, and Y. Vorobeychik, “A partially-supervised reinforcement learning framework for visual active search,” in *Neural Information Processing Systems (NeurIPS)*, Dec. 2023. arXiv: [2310.09689](https://arxiv.org/abs/2310.09689).
- [31] S. Khanal, S. Sastry, A. Dhakal, and N. Jacobs, “Learning tri-modal embeddings for zero-shot soundscape mapping,” in *British Machine Vision Conference (BMVC)*, Nov. 2023. arXiv: [2309.10667](https://arxiv.org/abs/2309.10667) [cs.CV].
- [32] Z. Xiong, F. Qiao, Y. Zhang, and N. Jacobs, “StereoFlowGAN: Co-training for stereo and flow with unsupervised domain adaptation,” in *British Machine Vision Conference (BMVC)*, Nov. 2023. arXiv: [2309.01842](https://arxiv.org/abs/2309.01842).
- [33] X. Xing, C. Peng, Y. Zhang, A.-L. Lin, and N. Jacobs, “AssocFormer: Association transformer for multi-label classification,” in *British Machine Vision Conference (BMVC)*, Nov. 2022.
- [34] E. Xing, X. Xing, L. Liu, N. Jacobs, Y. Qu, and G. Liang, “Neural network decision-making criteria consistency analysis via inputs sensitivity,” in *International Conference on Pattern Recognition (ICPR 2022)*, Aug. 2022. DOI: [10.1109/ICPR56361.2022.9956394](https://doi.org/10.1109/ICPR56361.2022.9956394)
- [35] S. Workman, M. U. Rafique, H. Blanton, and N. Jacobs, “Revisiting near/remote sensing with geospatial attention,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Acceptance rate: 25.33%, Jun. 2022. DOI: [10.1109/CVPR52688.2022.00182](https://doi.org/10.1109/CVPR52688.2022.00182)
- [36] X. Xing, G. Liang, Y. Zhang, S. Khanal, A.-L. Lin, and N. Jacobs, “ADViT: Vision transformer on multi-modality pet images for alzheimer disease diagnosis,” in *IEEE International Symposium on Biomedical Imaging (ISBI)*, Mar. 2022. DOI: [10.1109/ISBI52829.2022.9761584](https://doi.org/10.1109/ISBI52829.2022.9761584)
- [37] H. Blanton, S. Workman, and N. Jacobs, “A structure-aware method for direct pose estimation,” in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2022. DOI: [10.1109/WACV51458.2022.00028](https://doi.org/10.1109/WACV51458.2022.00028) arXiv: [2012.12360](https://arxiv.org/abs/2012.12360) [cs.CV].
- [38] Y. Zhang, G. Liang, and N. Jacobs, “Dynamic feature alignment for semi-supervised domain adaptation,” in *British Machine Vision Conference (BMVC)*, Nov. 2021. arXiv: [2110.09641](https://arxiv.org/abs/2110.09641).
- [39] G. Liang, X. Xing, L. Liu, Y. Zhang, Q. Ying, A.-L. Lin, and N. Jacobs, “Alzheimer’s disease classification using 2d convolutional neural networks,” in *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Jul. 2021. DOI: [10.1109/EMBC46164.2021.9629587](https://doi.org/10.1109/EMBC46164.2021.9629587)
- [40] Q. Ying, X. Xing, L. Liu, A.-L. Lin, N. Jacobs, and G. Liang, “Multi-modal data analysis for Alzheimer’s disease diagnosis: An ensemble model using imagery and genetic features,” in *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Jul. 2021. DOI: [10.1109/EMBC46164.2021.9630174](https://doi.org/10.1109/EMBC46164.2021.9630174)
- [41] Y. Zhang, G. Liang, Y. Su, and N. Jacobs, “Multi-branch attention networks for classifying galaxy clusters,” in *International Conference on Pattern Recognition (ICPR 2020)*, Acceptance rate: 28.47%, Jan. 2021. DOI: [10.1109/ICPR48806.2021.9412498](https://doi.org/10.1109/ICPR48806.2021.9412498)
- [42] G. Liang, Y. Zhang, X. Wang, and N. Jacobs, “Improved trainable calibration method for neural networks,” in *British Machine Vision Conference (BMVC)*, Sep. 2020.
- [43] M. U. Rafique, H. Blanton, N. Snively, and N. Jacobs, “Generative Appearance Flow: A hybrid approach for outdoor view synthesis,” in *British Machine Vision Conference (BMVC)*, Sep. 2020.

- [44] G. Liang, X. Wang, Y. Zhang, and N. Jacobs, “Weakly-supervised self-training for breast cancer localization,” in *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, (oral), Jul. 2020. DOI: [10.1109/EMBC44109.2020.9176617](https://doi.org/10.1109/EMBC44109.2020.9176617)
- [45] T. Salem, S. Workman, and N. Jacobs, “Learning a dynamic map of visual appearance,” in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Acceptance rate: 25%, Jun. 2020. DOI: [10.1109/CVPR42600.2020.01245](https://doi.org/10.1109/CVPR42600.2020.01245)
- [46] S. Workman and N. Jacobs, “Dynamic traffic modeling from overhead imagery,” in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Acceptance rate: 5.7% (oral), Jun. 2020. DOI: [10.1109/CVPR42600.2020.01233](https://doi.org/10.1109/CVPR42600.2020.01233)
- [47] G. Liang, X. Wang, Y. Zhang, X. Xing, H. Blanton, T. Salem, and N. Jacobs, “Joint 2d-3d breast cancer classification,” in *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Acceptance rate: 18% (oral), Nov. 2019. DOI: [10.1109/BIBM47256.2019.8983048](https://doi.org/10.1109/BIBM47256.2019.8983048) arXiv: [2002.12392](https://arxiv.org/abs/2002.12392).
- [48] Y. Zhang, X. Wang, H. Blanton, G. Liang, X. Xing, and N. Jacobs, “2d convolutional neural networks for 3d digital breast tomosynthesis classification,” in *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Acceptance rate: 18% (oral), Nov. 2019. DOI: [10.1109/BIBM47256.2019.8983097](https://doi.org/10.1109/BIBM47256.2019.8983097) arXiv: [2002.12314](https://arxiv.org/abs/2002.12314).
- [49] G. Liang, S. Fouladvand, J. Zhang, M. A. Brooks, N. Jacobs, and J. Chen, “GANai: Standardizing CT images using generative adversarial network with alternative improvement,” in *IEEE International Conference on Healthcare Informatics (ICHI)*, Jun. 2019. DOI: [10.1109/ICHI.2019.8904763](https://doi.org/10.1109/ICHI.2019.8904763)
- [50] Z. Bessinger and N. Jacobs, “A generative model of worldwide facial appearance,” in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, (oral), Jan. 2019. DOI: [10.1109/WACV.2019.00172](https://doi.org/10.1109/WACV.2019.00172)
- [51] N. Jacobs, A. Kraft, M. U. Rafique, and R. D. Sharma, “A weakly supervised approach for estimating spatial density functions from high-resolution satellite imagery,” in *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (ACM SIGSPATIAL)*, Acceptance rate: 22.5% (oral), Nov. 2018. DOI: [10.1145/3274895.3274934](https://doi.org/10.1145/3274895.3274934) arXiv: [1810.09528](https://arxiv.org/abs/1810.09528).
- [52] R. P. Mihail and N. Jacobs, “Automatic hand skeletal shape estimation from radiographs,” in *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Acceptance rate: 19.6%, Nov. 2018. DOI: [10.1109/BIBM.2018.8621196](https://doi.org/10.1109/BIBM.2018.8621196)
- [53] S. Schuler, M. Zhai, N. Jacobs, and M. Chandraker, “Learning to look around objects for top-view representations of outdoor scenes,” in *European Conference on Computer Vision (ECCV)*, Acceptance rate: 31.8%, Sep. 2018. DOI: [10.1007/978-3-030-01267-0_48](https://doi.org/10.1007/978-3-030-01267-0_48) arXiv: [1803.10870](https://arxiv.org/abs/1803.10870).
- [54] M. Zhai, T. Salem, C. Greenwell, S. Workman, R. Pless, and N. Jacobs, “Learning geo-temporal image features,” in *British Machine Vision Conference (BMVC)*, Acceptance rate: 29.5%, Sep. 2018.
- [55] C. Greenwell, S. Workman, and N. Jacobs, “What goes where: Predicting object distributions from above,” in *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, Jul. 2018. DOI: [10.1109/IGARSS.2018.8519251](https://doi.org/10.1109/IGARSS.2018.8519251) arXiv: [1808.00995](https://arxiv.org/abs/1808.00995).
- [56] T. Salem, M. Zhai, S. Workman, and N. Jacobs, “A multimodal approach to mapping soundscapes,” in *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, Jul. 2018. DOI: [10.1109/IGARSS.2018.8517977](https://doi.org/10.1109/IGARSS.2018.8517977)
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Workshop Papers

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- [49] P. Wang, S. Bhattacharyya, D. White, and N. Jacobs, “Visualization of Kentucky Lake,” in *Kentucky EPSCoR Conference*, Oct. 2011.

Technical Reports

- [1] A. Abrams, C. Hawley, K. Miskell, A. Stoica, N. Jacobs, and R. Pless, “Shadow estimation method for “the episolar constraint: Monocular shape from shadow correspondence”,” *arXiv*, vol. preprint 1304.4112 [cs.CV], Apr. 2013.
- [2] N. Jacobs, S. Schuh, and R. Pless, “On unusual pixel shapes and image motion,” Computer Science and Engineering, Washington University in St. Louis, MO, USA, Tech. Rep. WUCSE-2009-16, Jun. 2009.

Datasets

- [1] A. Abrams, J. Tucek, J. Little, N. Jacobs, and R. Pless, *LOST: Longterm observation of scenes (with tracks)*, <http://mvrl.github.io/LOST>.
- [2] M. T. Islam, C. Greenwell, and N. Jacobs, *GeoFaces: A large database of geolocated face patches*, <http://mvrl.github.io/GeoFaces>.
- [3] N. Jacobs, R. Pless, A. Abrams, and many others (see website for details), *AMOS: The archive of many outdoor scenes*, <https://mvrl.github.io/AMOS>.
- [4] P. Mihail, S. Workman, Z. Bessinger, and N. Jacobs, *SkyFinder: A large dataset of webcam images annotated with sky regions*, <https://mvrl.github.io/SkyFinder>.

- [5] M. U. Rafique, H. Blanton, and N. Jacobs, *Brooklyn Panorama Synthesis: A large dataset of panoramic images suitable for view synthesis evaluation*. <https://mvrl.github.io/GAF>.
- [6] T. Salem, S. Workman, M. Zhai, and N. Jacobs, *Cross-View Time (CVT)*, <https://mvrl.github.io/CVT>.
- [7] T. Salem, S. Workman, M. Zhai, and N. Jacobs, *Face2Year: A large number of images extracted from highschool yearbooks*, <https://mvrl.github.io/Face2Year>.
- [8] S. Workman and N. Jacobs, *Cross-View ScenicOrNot (CVSoN)*, <https://mvrl.github.io/CVSoN>.
- [9] S. Workman and N. Jacobs, *Crossview USA (CVUSA): A large dataset containing millions of pairs of ground-level and aerial/satellite images from across the United States*. <https://mvrl.github.io/CVUSA>.
- [10] S. Workman, M. Zhai, and N. Jacobs, *Horizon Lines in the Wild (HLW): A large database of images with known horizon-line location*, <http://mvrl.github.io/HLW>.

Patents

- [1] N. Jacobs and S. Workman, *Network architecture for generating a labeled overhead image*, US Patent App. 16/045,606, Jan. 2020.
- [2] J. A. G. Whitney, J. T. Fessler, Z. C. N. Kratzer, N. B. Jacobs, A. M. Whitney, et al., *Method and system for estimating error in predicted distance using RSSI signature*, Jan. 2016.

5 Talks

- 1. “From Social Media to Sanderlings: Multimodal Tools for Understanding and Mapping Biodiversity” (Keynote), Dec 2025, Workshop on Imageomics @ NeurIPS, San Diego, CA
- 2. “Learning to Map Anything, Anywhere, Anytime” (Sanghi College of Engineering Dean’s Special Invited Speaker), Dec 2025, Northern Arizona University, Flagstaff, AZ
- 3. “Learning to Map Anything, Anywhere, Anytime” (Invited Talk), Nov 2025, National Geospatial-Intelligence Agency, St. Louis, MO
- 4. “(Almost) Two Decades of Vision Across altitudes” (Keynote), Oct 2025, Workshop on 3D-VAST From street to space: 3D Vision Across altitudes (ICCV 2025 Workshop), Honolulu, Hawaii
- 5. “World Foundation Models for GeoAI” (Expert Panel), Sep 2025, Trillion Pixel Challenge Workshop, St. Louis, MO
- 6. “Toward Agentic GEOINT” (Lightning Talk), May 2025, GEOINT Symposium, St. Louis, MO
- 7. “Mapping and Multimodal Embeddings for Biodiversity” (Invited Talk), Mar 2025, Beyond Counting Trees (Bezos Earth Fund), Washington, DC
- 8. “Learning to Map Anything, Anywhere, Anytime” (Keynote), Nov 2023, The International Conference on Digital Image Computing: Techniques and Applications (DICTA), Port Macquarie, Australia
- 9. “Learning to Map Anything, Anywhere, Anytime” (Keynote), Oct 2023, ACM Multimedia Workshop (UAVs in Multimedia: Capturing the World from a New Perspective), Ottawa, Canada
- 10. “A Cost-Sensitive Approach To Dimensionality Reduction for Multispectral Imagery”, Jul 2023, International Geoscience and Remote Sensing Symposium, Pasadena, CA
- 11. “Explorations in Self-Supervised Learning for Change Detection”, Jul 2023, International Geoscience and Remote Sensing Symposium, Pasadena, CA

12. “Toward Dynamic Multimodal Remote Sensing: From Buildings and Populations to Soundscapes and Aesthetics”, Apr 2023, Living Earth Collaborative, Washington University, St. Louis, MO
13. “Domain-Inspired Deep Learning for Computer Vision, Remote Sensing, and Medical Imaging”, Oct 2022, Imaging Science Seminar, Washington University, St. Louis, MO
14. “Computer Vision for Multimodal Remote Sensing”, Aug 2022, WashU Geospatial Working Group Research Workshop, Washington University, St. Louis, MO
15. “A Structure-Aware Method for Direct Pose Estimation”, Jan 2022, IEEE Winter Conference on Applications of Computer Vision (WACV), Waikoloa Village, HI
16. Panelist for “Non-Traditional Careers in Computer Science” Nov 2021, ACM-W, University of Kentucky, Lexington, KY
17. “Mapping the Visual World Using Webcams, Cell Phones, and Satellites”, Oct 2021, Washington University in St. Louis, MO
18. “Learning Geo-Temporal Scene Models from Webcams, Cell Phones, and Satellites” (Keynote), Oct 2021, International Workshop on Distributed Smart Cameras, an ICCV Workshop (virtual)
19. “Mapping the Visual World Using Webcams, Cell Phones, and Satellites”, Dec 2020, University of Campinas, Unicamp, Brazil (virtual)
20. “Exploring the Intersection of Localization, Mapping, and Image Understanding” (Keynote), Aug 2020, ECCV Workshop on Long-Term Visual Localization (virtual)
21. “Deep Convolutional Neural Networks: Foundations to Frontiers (a 2-day short course)”, Mar 2020, Brazilian Space Agency (INPE), Sao Jose dos Campos, Brazil
22. “What, Where, and When: Mapping the World Using Webcams, Cell Phones, and Satellites”, Mar 2020, Brazilian Space Agency (INPE), Sao Jose dos Campos, Brazil
23. “Learning to Map Visual Appearance”, Feb 2020, Keeping Current Seminar, University of Kentucky (Computer Science), Lexington, KY
24. “Learning to Map Visual Appearance”, Jan 2020, Wageningen University, Netherlands
25. “What, Where, and When: Mapping the World Using Webcams, Cell Phones, and Satellites”, Nov 2019, University of Kentucky (Forestry), Lexington, KY
26. “Learning to Map the Visual World”, Jul 2019, Wright State University, Dayton, OH
27. “Understanding Places Using Ground-Level and Overhead Views” (Keynote), May 2019, Kentucky Geological Society (Annual Symposium), Lexington, KY
28. “Understanding Places Using Ground-Level and Overhead Views”, Feb 2019, Notre Dame University, South Bend, IN
29. “A Generative Model of Worldwide Facial Appearance” (Keynote), Jan 2019, Workshop on Demographic Variations in Performance of Biometric Algorithms, Waikoloa Village, HI
30. “A Generative Model of Worldwide Facial Appearance”, Jan 2019, IEEE Winter Conference on Applications of Computer Vision, Waikoloa Village, HI
31. “A Weakly Supervised Approach for Estimating Spatial Density Functions from High-Resolution Satellite Imagery”, Nov 2018, ACM SIGSPATIAL, Seattle, WA

32. “Understanding Places Using Ground-Level and Overhead Views”, Oct 2018, Commonwealth Computational Summit, Lexington, KY
33. “GeoLookbook: Modeling Worldwide Human Visual Appearance (Year 4)”, Sep 2018, National Academy of Sciences (IC Academic Research Symposium), Washington, DC
34. “Understanding Places Using Ground-Level and Overhead Views”, Aug 2018, Oak Ridge National Lab, Oak Ridge, TN
35. “WhatGoesWhere: Predicting Object Distributions from Above”, Jul 2018, IGARSS, Valencia, Spain
36. “Building World Models for Situated Training and Planning”, May 2018, Air Force Science and Technology 2030 Workshop, Bloomington, IN
37. “Recent Advances in Image Understanding”, May 2018, DASC, Lexington, KY
38. “(Tutorial) Recent Advances in Deep Learning: Fusing Overhead and Ground-Level Views for Remote Sensing”, April 2018, USGIF Annual Symposium, Tampa, FL
39. “Understanding Places Using Ground-Level and Overhead Views”, Feb 2018, CVPR Area Chair Meeting, Toronto, Canada
40. “GeoLookbook: Modeling Worldwide Human Visual Appearance (Year 3)”, Sep 2017, National Academy of Sciences (IC Academic Research Symposium), Washington, DC
41. “GPU Accelerated Computer Vision, Remote Sensing, and Machine Learning”, Aug 2017, Kentucky Geological Service, Lexington, KY
42. “Fusing Overhead and Ground-Level Imagery to Improve Scene Understanding”, Jul 2017, Planet, San Francisco, CA
43. “Learning about When and Where from Imagery”, Jun 2017, Orbital Insight, Mountain View, CA
44. “(Tutorial) Recent Advances in Deep Learning: Fusing Overhead and Ground-Level Views for Remote Sensing”, Jun 2017, USGIF Annual Symposium, San Antonio, TX
45. “How Computers See People (extended)”, May 2017, CCTS Biomedical Informatics Seminar Series, Lexington, KY
46. “Understanding Places Using Ground-Level and Overhead Views”, May 2017, Midwest Vision Meeting, Chicago, IL
47. “How Computers See People”, Feb 2017, Suds’n’Science Speaker Series, West Sixth Brewing, Lexington, KY
48. “Learning about When and Where from Imagery”, Feb 2017, University of Missouri, Department of Computer Science
49. “Localization, Mapping, and Image Understanding”, Feb 2017, USGIF Machine Learning Symposium
50. “Deep Convolutional Neural Networks: Concepts and Examples (in Computer Vision)”, Nov 2016, University of Kentucky, Society of Industrial and Applied Mathematics
51. “Crossview Convolutional Networks”, Oct 2016, Applied Imagery and Pattern Recognition, Washington, D.C.
52. “GeoLookbook: Modeling Worldwide Human Visual Appearance (Year 2)”, Sep 2016, National Academy of Sciences (IC Academic Research Symposium), Washington, DC
53. “Deep Convolutional Neural Networks: Concepts and Examples”, Jul 2016, University of Kentucky: Systems Biology and Omics Integration Seminar

54. “Crossview Methods for Localization and Mapping”, Jun 2016, IEEE CVPR Workshop on “Vision from Satellite to Street” (invited talk)
55. “A Fast Method for Estimating Transient Scene Properties”, Mar 2016, Winter Conference on Applications of Computer Vision, Lake Placid, NY
56. “Novel Cues for Geocalibration”, Feb 2016, Indiana University, Bloomington, IN
57. “Novel Cues for Camera Geocalibration”, Jan 2016, Uber Advanced Technology Center, Pittsburgh, PA
58. “Novel Cues for Geocalibration: Cloudy Days, Rainbows, and More”, Oct 2015, Carnegie Mellon University, Pittsburgh, PA
59. “Using Geotagged Internet Imagery to Understand the World”, Sep 2015, Université Laval, Quebec City, Canada
60. “face2gps: Estimating Geographic Location from Facial Features”, Sep 2015, International Conference on Image Processing, Quebec City, Canada
61. “GeoLookbook: Modeling Worldwide Human Visual Appearance”, Sep 2015, National Academy of Sciences (IC Academic Research Symposium), Washington, DC
62. “Exploring the Geo-Dependence of Human Face Appearance”, Mar 2014, Winter Conference on Applications of Computer Vision, Steamboat Springs, CO
63. “Estimating Cloudmaps from Outdoor Image Sequences”, Mar 2014, Winter Conference on Applications of Computer Vision, Steamboat Springs, CO
64. “Scene Geometry from Several Partly Cloudy Days”, Oct 2013, International Conference on Distributed Smart Cameras, Palm Springs, CA
65. “Unlocking the Potential of the Global Network of Outdoor Webcams”, Apr 2013, Rochester Institute of Technology
66. “Geo-temporal Computer Vision: Applications to the NGA”, Nov 2011, National Geospatial-Intelligence Agency
67. “Geo-temporal Computer Vision: Applications to the Army”, Oct 2011, Army Research Lab
68. “Localizing, Calibrating, and Using Thousands of Outdoor Webcams”, Feb 2011, University of North Carolina–Charlotte
69. “Using Clouds Shadows to Infer Scene Structure and Camera Calibration”, Jun 2010, CVPR, San Francisco, CA
70. “Passive Vision and The Power of Collective Imaging”, Apr 2010, Object Video Inc., Reston, VA
71. “Localizing, Calibrating, and Using Thousands of Outdoor Webcams”, Apr 2010, University of Kentucky
72. “Time-Lapse Vision: Localizing, Calibrating, and Using Thousands Outdoor Webcams”, Apr 2010, Google, Mountain View, CA
73. “Passive Vision and The Power of Collective Imaging”, Jan 2010, Google, Mountain View, CA
74. “Incorporating Domain Constraints in Urban Vehicle Tracking”, Nov 2010, University of Missouri, Columbia, MO
75. “Compressive Sensing and Differential Image-Motion Estimation”, Mar 2010, ICASSP, Dallas, TX
76. “The Global Network of Outdoor Webcams: Properties and Applications ”, Nov 2009, ACM GIS, Seattle, WA

77. “Passive Vision: The Global Webcam Imaging Network”, Oct 2009, AIPR, Washington, DC
78. “Calibrating and Using the Global Network of Outdoor Webcams”, Aug 2009, ICDSC, Italy
79. “Adventures in Archiving and Using Three Years of Webcam Images”, Jun 2009, CVPR Workshop on Internet Vision, Miami, FL
80. “Recent Work: Webcams and Grooves”, Aug 2009, Object Video, Reston, VA
81. “Location-Specific Models for Tracking”, Jan 2008, WMVC, Copper Mountain, CO
82. “Using natural cues to geo-locate and geo-orient distributed cameras”, Jan 2008, VISN, Copper Mountain, CO
83. “Foreground Modeling: The Shape of Things That Came”, Feb 2007, WMVC, Austin, Texas

6 Service

University Service

- Washington University in St. Louis (2022–present)
 - 2022–present: Co-Director, Geospatial Research Initiative
 - 2022–present: Research Council, Taylor Geospatial Institute
 - 2023–present: Director of PhD Admissions, Department of Computer Science & Engineering
 - 2023–present: Faculty Technology Advisory Committee, James McKelvey School of Engineering
 - 2023–present: Curriculum Committee, Imaging Science Program
 - 2023–present: Faculty Advisor, WashU Robomaster Club
 - F2024: Faculty Search Committee, Imaging Science Program
 - F2023, F2024: Faculty Search Committee, Computer Science & Engineering Department
 - 2022–2023: Leadership Team, Geospatial Working Group
 - 2022–2023: Strategic Planning Steering Committee, James McKelvey School of Engineering
- University of Kentucky (2010–2022)
 - 2021–2022: Institute for Biomedical Informatics: Steering Committee
 - 2019–2022: Computer Science Department: Executive Committee
 - 2019–2022: College of Engineering: Master Planning/Space Committee
 - 2018–2019, 2020–2022: College of Engineering: Research Advisory Committee
 - 2020–2022: College of Engineering: Graduate Studies Team
 - 2013–2017, 2018–2022: Computer Science Department: Faculty Search Committee
 - 2020–2021: Computer Science Department: Chair Search Committee
 - 2020: College of Engineering: Recruiting Advisory Committee
 - 2018–2019: University Senate (Academic Facilities Committee, Technology Committee)
 - 2017: Member (Information Technology Task Force for Research Enablement and Outreach)
 - 2015–2016: Computer Science Department: ABET Committee
 - 2010–2012, 2015–2016: Computer Science Department: Media and Outreach
 - 2013: Center for Visualization and Virtual Environment: Director Search Committee
 - 2013: Computer Science Department: Chair Search Committee
 - 2012–2013: Computer Science Department: Curriculum Development Committee
 - 2012–2013, 2015: Engineering Day (oral presentation and/or software demonstration)

Professional Service

- Area Chair:
 - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2018, 2019, 2021, 2023, 2024 (senior), 2025 (lead), 2026–2027 (senior)]
 - European Conference on Computer Vision (ECCV) [2022, 2024 (lead)]
 - IEEE/CVF International Conference on Computer Vision (ICCV) [2023, 2025 (lead)]
 - Conference on Neural Information Processing Systems (NeurIPS) [2024]
 - IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2014, 2022, 2023, 2027 (senior)]
- Organizing Committees:
 - IEEE/ISPRS Workshop on Large Scale Computer Vision for Remote Sensing Imagery (EARTHVISION) [2019–2023, 2025–2026]
 - Foundational Models Beyond the Visual Spectrum (workshop at IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)) [2026]
 - Doctoral Consortium Co-Chair: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2017, 2019, 2024, 2025]
 - Industrial/Government Relations Chair: IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2024]
 - Doctoral Consortium Chair: IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2018, 2022]
 - Video Proceedings Chair: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2015]
 - IEEE Workshop on Motion and Video Computing (WMVC) [2011]
- Guest Editor:
 - ISPRS Journal of Photogrammetry and Remote Sensing [2024], Special Issue “Vision Language Models for Remote Sensing Analysis and Interpretation”
 - IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (J-STARS) [2021], Special Issue “Integrating User Generated Contents for Remote Sensing Applications”
 - Elsevier Computer Vision and Image Understanding (CVIU) [2019], Special Issue “Computer Vision for Remote Sensing”
- Panelist:
 - Roundtable Discussion at 1st Workshop on Computer Vision for Earth Observation (CV4EO) Applications 2024 (Hosted as part of IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2024)
- Session Chair:
 - IEEE International Geoscience and Remote Sensing Symposium (IGARSS) [2020,2023]
 - IEEE/ISPRS Workshop on Large Scale Computer Vision for Remote Sensing Imagery (EARTHVISION) [2019]
 - IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2016, 2019, 2022, 2024]
 - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2018, 2024, 2025]

- IEEE/ACM International Conference on Distributed Smart Cameras [2013]
- Reviewing for Journals:
 - Proceedings of the National Academy of Sciences [2024]
 - IEEE Transactions on Geoscience and Remote Sensing [2017, 2020]
 - ISPRS Journal of Photogrammetry and Remote Sensing [2019, 2020]
 - IEEE Transactions on Pattern Analysis and Machine Intelligence [2011×2, 2012, 2018, 2019]
 - IEEE Transactions on Multimedia [2011, 2016]
 - Elsevier Computer Vision and Image Understanding [2010, 2013, 2016×2]
 - IEEE Transactions on Computational Imaging [2016]
 - IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (J-STARS) [2015]
 - Springer Machine Vision and Applications [2014]
 - IEEE Sensors [2014]
 - Elsevier Image and Vision Computing [2013]
 - IEEE Transactions on Circuits and Systems for Video Technology [2007–2011]
 - IEEE Computer Graphics and Applications [2010]
 - IEEE Transactions on Aerospace and Electronic Systems [2010]
 - Elsevier Computers and Electronics in Agriculture [2010]
 - Cartography and Geographic Information Science [2010]
- Program Committee / Reviewer for:
 - Conferences
 - * IACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR) [2026]
 - * International Conference on Learning Representations (ICLR) [2022, 2025]
 - * IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2021, 2024, 2025]
 - * IEEE International Geoscience and Remote Sensing Symposium (IGARSS) [2020]
 - * British Machine Vision Conference (BMVC) [2020]
 - * IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2006–2017, 2020]
 - * European Conference on Computer Vision (ECCV) [2010, 2014, 2020]
 - * Neural Information Processing Systems (NeurIPS) [2010–2012, 2020]
 - * AAAI Conference on Artificial Intelligence (AAAI) [2020, 2026 (social impact track)]
 - * IEEE/CVF International Conference on Computer Vision (ICCV) [2007, 2009, 2019, 2021]
 - * Asian Conference on Computer Vision (ACCV) [2010, 2016]
 - * IEEE International Conference on Robotics and Automation (ICRA) [2016]
 - * International Conference on Machine Learning (ICML) [2012]
 - * IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS) [2010]
 - Workshops
 - * ECCV Workshop TerraBytes — Towards global datasets and models for Earth observation [2026]
 - * ICLR Workshop on Machine Learning for Remote Sensing [2023]
 - * IEEE/ISPRS Workshop on Large Scale Computer Vision for Remote Sensing Imagery (EARTHVISION) [2017, 2019, 2020]
 - * CVPR Workshop on Photogrammetric Computer Vision [2019]

- * CVPR Workshop on DeepGlobe Satellite Challenge [2018]
 - * ACM International Workshop on Geotagging and Its Applications [2013]
 - * ICCV Workshop on Computer Vision for Converging Perspectives [2013]
 - * IEEE Workshop on Applications of Computer Vision [2012–2013]
 - * ECCV Workshop on Visual Analysis and Geo-Localization of Large-Scale Imagery [2012]
 - * ACM Workshop on Geotagging and Its Applications in Multimedia [2012]
 - * IEEE Workshop on Motion and Video Computation [2009–2011]
- Reviewing for Funding Agencies:
 - Panelist for NSF Information and Intelligent Systems Division [2015, 2016, 2017, 2019]
 - Panelist for NSF Division of Industrial Innovation and Partnerships [2016]
 - External reviewer for NSF Information and Intelligent Systems Division [2015]
 - External reviewer for Fonds de recherche du Quebec [2014]
 - Advisory Committee for:
 - 1st Workshop on Computer Vision for Earth Observation (CV4EO) Applications 2024 (Hosted as part of IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2024)

Memberships

- Senior Member: Institute of Electrical and Electronics Engineers
- Full Member: British Machine Vision Association and Society for Pattern Recognition
- Affiliate Member: International Association of Pattern Recognition

Graduate Committee Service

Ph.D. Committees

- | | |
|--|--|
| 1. David Sundarsingh (Electrical and Systems Engineering, TBD) | 12. Siyuan Shen (Energy, Environmental & Chemical Engineering, 2026) |
| 2. Timur Kasimov (Computer Science, TBD) | 13. Ruiqi Wang (Computer Science, 2026) |
| 3. Yuanlong Su (EECE, TBD) | 14. Tri Pham (Computer Science, 2025) |
| 4. Chongsong Huang (Computer Science, TBD) | 15. Sheng-Chieh Lin (Physics, 2025) |
| 5. Ben Warner (Computer Science, TBD) | 16. Weijie Gan (Computer Science, 2025) |
| 6. Evelyn Yang (Computer Science, TBD) | 17. Yiwen Ju (Computer Science, 2025) |
| 7. Yu Yan (EECE, TBD) | 18. Hao Liu (Computer Science, 2025) |
| 8. Lisa Liao (Computer Science, TBD) | 19. Junlin Wu (Computer Science, 2025) |
| 9. Anindya Sarkar (Computer Science, TBD) | 20. Gustavo Gratacós (Computer Science, 2024) |
| 10. Nischal Khanal (Imaging Science, TBD) | 21. Ye Htet (Computer Science, 2024) |
| 11. Lydia Reader (2026, Computational & Data Sciences) | 22. Minoos Hosseinzadeh (Computer Science, 2024) |
| | 23. Stephen Parsons (Computer Science, 2023) |

24. Md Selim (Computer Science, 2023)
25. Ashutosh Timilsina (Computer Science, 2023)
26. Arnab Sarkar (Physics, 2022)
27. David Adeniji (Mechanical Engineering, 2022)
28. Tarannum Shaila Zaman (Computer Science, 2022)
29. Chengxi Li (Computer Science, 2022)
30. Sajad Javadinasab Hormozabad (Civil Engineering, 2021)
31. Alireza Shirvani (Computer Science, 2021)
32. Narjes Bozorg (Electrical Engineering, 2020)
33. Kyle Helfrich (Math, 2020)
34. Sifei Han (Computer Science, 2019)
35. Xinxin Zuo (Computer Science, 2019)
36. Jinping Zhuge (Math, 2019)
37. Nkiruka Uzuegbunam (Electrical Engineering, 2018)
38. Anthony Rios (Computer Science, 2018)
39. Po-Chang Su (Electrical Engineering, 2017)
40. Yajie Zhao (Computer Science, 2017)
41. Wesley Hough (Computer Science, 2017)
42. Hasan Sajid (Electrical Engineering, 2016)
43. Harikrishnan Unnikrishnan (Electrical Engineering, 2015)
44. Bo Fu (Computer Science, 2015)
45. Shaoceng Wei (Statistics, 2015)
46. Yan Huang (Computer Science, 2014)
47. Mao Ye (Computer Science, 2014)
48. Chenxi Zhang (Computer Science, 2014)
49. Ju Shen (Electrical Engineering, 2014)

External Ph.D. Committees

1. Rachel Badzioch (Notre Dame University, Biological Sciences, TBD)
2. Christoph Gerhardt (Technische Universität Ilmenau, Computer Science, TBD)
3. Morris Alper (Tel Aviv University, Computer Science, 2025)
4. Eike Jens Hoffmann (Technical University of Munich, Data Science in Earth Observation, 2022)
5. Ahmed Nassar (IRISA, Université Bretagne Sud, Vannes, Computer Science, 2021)
6. Céline Portenier (University of Bern, Computer Science, 2021)
7. Raian Maretto (National Institute for Space Research, Geoinformation Science, 2020)
8. Shivangi Srivastava (Wageningen University, Netherlands, Computer Science, 2020)
9. Nam Vo (Georgia Institute of Technology, Computer Science, 2019)
10. Yannick Hold-Geoffroy (Laval University, Quebec, CA, Computer Science, 2018)
11. Ethan Welty (University of Colorado–Boulder, Environmental Studies, 2018)

Master's Committees

1. Yiding Yang (Computer Science, 2026)
2. Aadarsha Gopala Reddy (Computer Science, 2026)
3. Aaron Luo (Computer Science, 2026)
4. Stuart Aldrich (Computer Science, 2026)
5. Weining Wang (Computer Science, 2026)
6. Kyle Wolford (Computer Science, 2025)
7. Haris Naveed (Computer Science, chair, 2025)
8. Yuxuan Yang (Computer Science, 2025)
9. Zifan Wang (Computer Science, 2025)
10. Sizhe Zhang (Statistics, 2024)

11. Yin Li (Computer Science, 2024)
12. Evin Jaff (Computer Science, 2024)
13. Owen Ma (Computer Science, 2024)
14. Patrick Lynch (Computer Science, 2024)
15. Joshua Tang (Computer Science, 2024)
16. Kyle Montgomery (Computer Science, 2024)
17. Fiona Xu (Computer Science, 2023)
18. Emma McMillian (Computer Science, 2023)
19. Chang Ti (Computer Science, 2023)
20. Brian Chao (Computer Science, 2023)
21. Yihang Xu (Computer Science, 2023)
22. David Sarpong (Computer Science, 2023)
23. Peizhen Tong (Computer Science, 2023)
24. Di Huang (Computer Science, 2023)
25. Nan Huang (Computer Science, 2023)
26. Zihao Zou (Computer Science, 2022)
27. Zhou Chu (Computer Science, 2022)
28. Aiden McIlraith (Computer Science, 2022)
29. Yuan Liu (Computer Science, 2022)
30. Subash Khanal (Electrical Engineering, 2020)
31. Genghis Goodman (Computer Science, 2019)
32. Jonathan Dingess (Computer Science, 2019)
33. Ryan Zembrodt (Computer Science, 2019)
34. Xiaofei Zhang (Computer Science, 2017)
35. Qingguo Xu (Computer Science, 2017)
36. DhiShankar Bhattacharya (Computer Science, 2017)
37. Stanley Rosenbaum (Computer Science, 2016)
38. Sean Karlage (Computer Science, 2016)
39. Hasan Sajid (Electrical Engineering, 2014)
40. Edwin Prem Kumar Sathiyamoorthy (Electrical Engineering, 2011)

7 Teaching

Courses Taught

The following list summarizes the traditional, classroom courses I have taught:

- *Advances in Computer Vision*, CSE 659a/5519, [F2024,F2025], Washington University in St. Louis
- *Computer Vision*, CSE 559a/5509, [S2023, S2024, S2025, S2026], Washington University in St. Louis
- *Machine Learning*, CS 460g, [F2012, F2013, F2014, F2016, F2018, F2019], University of Kentucky
- *Computer Vision*, CS 636, [S2011, S2013, S2017], University of Kentucky
- *Learning-Based Methods for Computer Vision*, CS 585/685, [S2015], University of Kentucky
- *Advanced Topics in Computer Science: Machine Learning*, CS 685, [S2012], University of Kentucky
- *Intermediate Topics in Computer Science: Computational Photography*, CS 585, [F2010, F2011], University of Kentucky
- *Theory of Computation*, CECS 341, [F2002], University of Missouri

8 Lab Members / Mentees

Postdoctoral Scholars

1. [Subash Khanal](#) Dates: 2025–present
Research Focus: Geospatial Representation Learning, Remote Sensing
2. [Adeel Ahmad](#) (Ph.D. Geomatics, University of Punjab–Lahore) Dates: 2023–2025
Research Focus: Remote Sensing, Deep Learning, Land-Use Modeling
3. [Benjamin Brodie](#) (Ph.D. Mathematics, University of Kentucky) Dates: 2020–2022
Research Focus: Change Detection, Object Tracking, Re-Identification, Metric Learning

Research Staff

1. [Hamza Iqbal](#) (MS, Computer Science, Washington University in St. Louis) Dates: 2026–present
Role: Research Associate
2. [Jonathan Lin](#) (MS, Computer Science, Washington University in St. Louis) Dates: 2024–present
Role: Research Associate

Ph.D. Students

1. [Chih-Chuan \(Leo\) Hsu](#) Degree: Ph.D., Computer Science
Title: TBD Date: TBD
2. [Brian Wei](#) Degree: Ph.D., Computer Science
Title: TBD Date: TBD
3. [Danni Beaulieu](#) Degree: Ph.D., Computer Science
Title: TBD Date: TBD
4. [Tianyi Gao](#) Degree: Ph.D., Computer Science
Title: TBD Date: TBD
5. [Feng Qiao](#) Degree: Ph.D., Computer Science
Title: TBD Date: TBD
6. [Carlos Lopez de la Cerda Bazaldu](#) [*co-chair* with Jacob Montgomery] Degree: Ph.D., Computational & Data Sciences
Title: TBD Date: TBD
7. [Daniel Cher](#) Degree: Ph.D., Computational & Data Sciences
Title: TBD Date: TBD
8. [Eric Xing](#) Degree: Ph.D., Computer Science
Title: TBD Date: TBD
9. Tong Li [*co-chair* with Joshua Oltmanns] Degree: Ph.D., Computational & Data Sciences
Title: TBD Date: TBD
10. [Nia Hodges](#) Degree: Ph.D., Systems Science and Mathematics
Title: TBD Date: TBD
11. [Alex Wollam](#) Degree: Ph.D., Computer Science
Title: TBD Date: TBD

12. [Zhexiao Xiong](#) Degree: Ph.D., Computer Science
Title: TBD Date: May 2027 (est)
13. Michael Lanier [*co-chair* with Yevgeniy Vorobeychik] Degree: Ph.D., Computer Science
Title: TBD Date: May 2027 (est)
14. [Srikumar Sastry](#) Degree: Ph.D., Imaging Science
Title: Task-Aligned Multimodal Representation Learning Date: Dec 2026 (est)
15. [Aayush Dhakal](#) Degree: Ph.D., Computer Science
Title: TBD Date: Dec 2026 (est)
16. [Subash Khanal](#) Degree: Ph.D., Computer Science
Title: Multimodal Representation Learning for Geospatial Soundscape Mapping Date: Jul 2025
Employment: Postdoctoral Scholar, Washington University in St. Louis
17. [Xin Xing](#) [*co-chair* with Ai-Ling Lin] Degree: Ph.D., Computer Science
Title: Structured Attention for Image Analysis Date: Nov 2023
Employment: Assistant Professor, University of Nebraska-Omaha
18. [Yu Zhang](#) Degree: Ph.D., Computer Science
Title: Multimodal Domain Generalization Date: Mar 2023
Employment: Assistant Professor, Computer Science, Boise State University
19. [Connor Greenwell](#) Degree: Ph.D., Computer Science
Title: Probabilistic Cross-Domain Representation Learning Date: Jun 2022
Employment: Senior R&D Engineer, Kitware Inc.
20. [Hunter Blanton](#) Degree: Ph.D., Computer Science
Title: Revisiting Absolute Pose Regression Date: Aug 2021
Employment: Senior Computer Vision Engineer, Yembo (startup)
21. [Usman Rafique](#) [*co-chair* with Samson Cheung] Degree: Ph.D., Electrical Engineering
Title: Weakly Supervised Learning for Multi-Image Synthesis Date: Jul 2021
Employment: Research Scientist, Kitware Inc.
22. [Gongbo “Tony” Liang](#) Degree: Ph.D., Computer Science
Title: Clinical-Inspired Multi-Modal Deep Learning Medical Imaging Analysis Date: Oct 2020
Employment: Assistant Professor, Computer Science, Eastern Kentucky University
23. [Tawfiq Salem](#) Degree: Ph.D., Computer Science
Title: Learning to Map the Visual and Auditory World Date: Jul 2019
Employment: Visiting Assistant Professor, Computer and Information Technology, Purdue University
24. [Zach Bessinger](#) Degree: Ph.D., Computer Science
Title: Modeling and Mapping Location-Dependent Human Appearance Date: Dec 2018
Employment: Senior Applied Scientist, Zillow
25. [Menghua “Ted” Zhai](#) Degree: Ph.D., Computer Science
Title: Deep Probabilistic Models for Camera Geo-Calibration Date: Dec 2018
Employment: Computer Vision Engineer, MatrixTime (startup)
26. [Scott Workman](#) Degree: Ph.D., Computer Science
Title: Leveraging Overhead Imagery for Localization, Mapping, and Understanding Date: Apr 2018
Employment: Research Scientist, DZYNE Technologies

27. [Hamid Hamraz](#) Degree: Ph.D., Computer Science
 Title: Computational Forest Modeling using Airborne Remote Sensing LiDAR Date: Apr 2018
 Employment: Computational and Data Scientist, Microsoft
28. [Mohammad T. Islam](#) Degree: Ph.D., Computer Science
 Title: Analyzing the Geo-Dependence of Human Face Appearance and Its Applications Date: Jul 2016
 Employment: Associate Professor, Computer Science, Southern Connecticut State University
29. [Paul Mihail](#) [*co-chair* with Judy Goldsmith] Degree: Ph.D., Computer Science
 Title: Visualizing and Predicting the Effects of Rheumatoid Arthritis on Hands Date: May 2014
 Employment: Professor, Computer Science, Valdosta State University

Visiting Scholars

1. [Abby Stylianou](#) (Saint Louis University) [*Visiting Associate Professor*] Dates: 2025–2026
 Topic: TBD
2. [Clemens Mosig](#) (Leipzig University) Dates: 2025
 Topic: Deep Learning for Tree Mortality Detection
3. [José Nascimento](#) (UNICAMP, Brazil) Dates: 2024
 Topic: TBD
4. [Alex Levering](#) (Wageningen University) Dates: 2022–2023
 Topic: Landscape Quality Assessment
5. [Rafael Padilha](#) (UNICAMP, Brazil) Dates: 2019–2020
 Topic: Image Forensics
6. [Raian Vargas Maretto](#) (INPE, Brazil) Dates: 2018–2019
 Topic: Deforestation Detection
7. [Patrick Tutzauer](#) (University of Stuttgart) Dates: Fall 2017
 Topic: Geospatial Trajectory Modeling

Master's Students

1. Helen Lei Degree: MS, Computer Science
 Title: Unified Multimodal Learning for Cross View Geospatial Understanding and Generation Date: Fall 2026
2. Rukai Gao Degree: MS, Computer Science
 Title: 3D Laryngeal Tumor Mapping via Near-Field SLAM Date: Fall 2026
3. Mohammad Rouie Miab Degree: MS, Computer Science
 Title: Improving Semantic Precision in Text-to-Image Diffusion Models via Latent-Space Optimization and Semantically-Parsed Evaluation Date: Spring 2026
4. Haris Naveed Degree: MS, Computer Science
 Title: Deep Learning for Tree Canopy Height Estimation Date: Spring 2025
5. Jackson McCall Degree: MS, Computer Science
 Title: GeoSynth++: Large-Scale Satellite Image Synthesis Date: Spring 2025
6. Jingyun Ma Degree: MS, Computer Science
 Title: Large Language Models for MCNP Input Generation Date: Spring 2025

7. Shuhan (Steven) Zhang Degree: MS, Computer Science
Title: Query-Specific Feature Transformation for Fine-Grained Image Retrieval Date: Spring 2025
8. Vinh Pham Degree: MS, Engineering Data Analytics & Statistics
Title: Automated Satellite Imagery Analysis for Global Agricultural Field Boundary Detection Date: Spring 2025
9. Ethan Weilheimer Degree: MS (thesis), Computer Science
Title: Partially Supervised Reinforcement Learning for GPS-Denied Navigation Date: Spring 2025
10. Myan Sudharsanan Degree: MS, Computer Engineering
Title: VisionLLMs in the Automotive Domain: Finegrained Details of Traffic Signs and Passenger Vehicles
Date: Spring 2025
11. Lunchi Guo Degree: MS, Computer Engineering
Title: MedVid-Align: Intelligent Analysis of Medical Procedure Videos Date: Spring 2025
12. Lunchi Guo Degree: MS, Computer Engineering
Title: Deep Learning-Based Coordinate Prediction from Medical Fluoroscopic Images for Improved Radiation
Date: Fall 2024
13. Wanzhou Liu Degree: MS, Computer Science
Title: High Efficiency Generalizable Driving World Model Date: Fall 2024
14. Nia Hodges Degree: MS, Engineering Data Analytics & Statistics
Title: Wide-Area Image Localization Date: May 2024
15. Hongzhang Wang Degree: MS, Computer Science
Title: Monocular Depth Estimation Date: Dec 2023
16. Alex Wollam Degree: MS, Computer Science
Title: Exploring Sequential Outdoor Panorama Synthesis with Diffusion Models Date: Jul 2023
17. Alex Greene Degree: MS, Computer Science
Title: Using Aerial Imagery to Estimate Ground-level Object Distributions Date: May 2023
18. Zachary Pulliam Degree: MS, Data Science
Title: TBD Date: 2022
19. Jacob Birge Degree: MS, Computer Science
Title: A Cost-Sensitive Approach To Multimodal Fusion Date: Dec 2021
20. [David T. Jones](#) Degree: MS, Computer Science
Title: Intensity Harmonization for Airborne LiDAR Date: May 2021
21. [Armin Hadzic](#) Degree: MS, Computer Science
Title: Estimating Free-Flow Speed with LiDAR and Overhead Imagery Date: May 2020
22. [Weilian “William” Song](#) Degree: MS, Computer Science
Title: Image-Based Roadway Assessment using Convolutional Neural Networks Date: May 2019
23. [William “Derek” Jones](#) [co-chair with Sally Ellingson] Degree: MS, Computer Science
Title: Scalable Feature Selection and Extraction with Applications in Kinase Polypharmacology Date: May 2018
24. [Ryan Baltenberger](#) Degree: MS, Computer Science
Title: Estimating Transient Scene Attributes Using Deep Convolutional Neural Networks Date: May 2016
25. [Feiyu Shi](#) Degree: MS, Electrical Engineering
Title: Principal Component Analysis For Multi-size Images Date: Dec 2013

Undergraduate Research Students

1. [Kexing \(Ariana\) Li](#) Dates: Fall 2025
Title: TBD
2. [Phoenix Jarosz](#) Dates: Fall 2025
Title: TBD
3. [Rowan Sandvig](#) Dates: Fall 2025
Title: TBD
4. [Nick Haisler](#) Dates: 2025 (REU, Drake University)
Title: TBD
5. Ice Cui Dates: Fall 2025
Title: Leveraging Diffusion Transformers to Improve GeoSynth Scalability and Cross-Scale Consistency
6. Dev Gupta Dates: Fall 2025
Title: Temporally-Consistent Stereo Video Generation
7. Brian Wei Dates: Fall 2025
Title: Stereo Video Analysis
8. Brian Song Dates: 2024 (REU, Vanderbilt University)
Title: TBD
9. [Gabriella Ramirez](#) Dates: 2024 (REU, University of Missouri)
Title: TBD
10. John Lim Dates: 2024 (REU, Northwestern University)
Title: TBD
11. Eddie Choi Dates: 2024 (Post-Bacc Research Assistant)
Title: TBD
12. [Daniel Khen](#) Dates: 2024
Title: TBD
13. Sam Schwartz Dates: 2023
Title: TBD
14. Josh Liu Dates: 2023
Title: TBD
15. [Cohen Archbold](#) Dates: 2020–2022
Title: Automatic Real-Estate Price Estimation
16. Matthew Mitchell Dates: 2022
Title: Remote Sensing for Social Good
17. Gareth Walker Dates: 2022
Title: Remote Sensing for Social Good
18. [Julia Stekardis](#) Dates: 2021–2022
Title: Large-Scale Image Geo-Localization
19. Evan Bolton Dates: 2021
Title: Generating Synthetic Training Data using a Game Engine

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| 20. Shashank Bhatt
Title: Multi-Object Tracking | Dates: 2019–2022 |
| 21. Nicole Wong
Title: TBD | Dates: 2021 |
| 22. Sean Grate
Title: Deep Learning for Point Clouds | Dates: 2019–2020 |
| 23. Evan Yang
Title: TBD | Dates: 2019 |
| 24. Thomas Barber
Title: Deep Learning for Remote Sensing | Dates: 2019 |
| 25. Yuhan Long
Title: Deep Learning for Medical Imaging | Dates: 2019 |
| 26. Weilian Song
Title: Applications of Deep Convolutional Neural Networks to Geometric Computer Vision | Dates: 2016–2019 |
| 27. Aaron Mueller
Title: Deep Learning for Educational Data | Dates: 2018 |
| 28. Sam Davidson
Title: Applications of Generative Adversarial Networks to Social Media Imagery | Dates: 2016–2017 |
| 29. Connor Greenwell
Title: Interactive Methods for Aerial Imagery Understanding | Dates: 2014–2016 |
| 30. Ryan Baltenberger
Title: Understanding Outdoor Scene Appearance | Dates: 2012–2015 |
| 31. Angelo Stekardis
Title: Understanding Facial Expressions | Dates: 2014–2015 |
| 32. J. David Smith
Title: User-in-the-loop Camera Calibration | Dates: 2013–2015 |
| 33. Noora Aljabi
Title: Using Flickr to Map Phenological Trends | Dates: 2013 |
| 34. Kyle Kolpek
Title: Aerial Image Registration | Dates: 2012 |
| 35. Jim Knochelmann
Title: User-Tools for Aerial Image Registration | Dates: 2011–2012 |

High School Research Students

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| 1. Krishna Bhatraju
Title: Deep Motion Estimation | Dates: 2021–2022 |
| 2. William Greenlee
Title: Deep Learning for Computer Vision | Dates: 2021–2022 |
| 3. Chris Wang
Title: Multimodal Medical Imaging for Alzheimer’s Disease Classification | Dates: 2019–2021 |

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| 4. Nicole Wong
Title: Learning-Based View Synthesis | Dates: 2019–2020 |
| 5. Whitney Dawson
Title: TBD | Dates: 2019 |
| 6. Cohen Archbold
Title: Photo-Geolocation using Convolutional Neural Networks | Dates: 2017–2018 |
| 7. Ryan Landry
Title: RRADCL: Rapid Roadway Assessment with Deep Convolutional Learning | Dates: 2017–2018 |
| 8. C. J. Labianca
Title: Evaluation of Optimization Algorithms for Deep Convolutional Neural Networks | Dates: 2016–2017 |
| 9. Andrew Albrecht
Title: Mapping Social Media Imagery | Dates: 2016–2017 |
| 10. Andrew Tapia
Title: Estimating Surface Reflectivity | Dates: 2014–2015 |
| 11. Alex Lucas
Title: Evaluation of Automatic Face Detection Methods | Dates: 2013–2014 |
| 12. Ryan Baltenberger
Title: Gesture-Based User Interaction with the Microsoft Kinect | Dates: 2011–2012 |